EXPLORATORY DATA ANALYSIS RESEARCH PAPER

Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park

An Empirical Study of Temporal Demand Patterns in Public Ferry Ticketing Systems

Prepared using Python (Pandas, NumPy, Matplotlib, Seaborn) and MySQL
Dataset: Toronto Island Ferry Tickets — 261,538 transaction records
Date: July 2026

Abstract

This research paper presents a comprehensive exploratory data analysis (EDA) of the Toronto Island Ferry Tickets dataset, comprising 261,538 time-stamped transaction records capturing ferry ticket sales and redemption activity. The primary objective of this study is to identify temporal demand patterns — hourly, daily, and seasonal — governing ferry ticket sales and redemptions, and to translate these patterns into actionable operational insights. The dataset was ingested, cleaned, type-corrected, and enriched with derived temporal features (hour of day, day name, month, season, weekend/weekday classification, day-period bucket, and a sales-redemption gap metric) before being persisted to a MySQL relational database for downstream analysis. Using Python's data science stack (Pandas, NumPy, Matplotlib, and Seaborn), the analysis reveals that ticket demand is highly concentrated in the afternoon hours (12:00 PM to 3:00 PM), that weekends — particularly Saturday and Sunday — account for a disproportionately large share of total ticket volume, and that Summer is the peak season by a wide margin while Winter represents the off-peak period. Furthermore, sales and redemption counts exhibit a close statistical correspondence across the observation period (50.4% versus 49.6% of combined volume, a difference of only 0.8 percentage points), suggesting an operationally efficient ticketing process with a low rate of ticket non-redemption. Rolling-average smoothing confirms a single, well-defined midday demand peak rather than multiple demand spikes throughout the day. These findings carry direct implications for staffing, ferry scheduling, and capacity planning at the Toronto Island Park terminal.

Keywords: Exploratory Data Analysis, Time Series, Ferry Ticketing, Demand Forecasting, Seasonal Analysis, Toronto Island Park, Pandas, Data Visualization.

1. Introduction

Public ferry services that operate on fixed schedules face a recurring operational challenge: matching vessel capacity and staffing levels to fluctuating, time-dependent passenger demand. The Toronto Island Park ferry system, which transports visitors between mainland Toronto and the Toronto Islands, generates a continuous stream of ticket sales and redemption events that can be mined to understand how demand varies across hours of the day, days of the week, and seasons of the year.

This study undertakes an end-to-end exploratory data analysis of a real-world ferry ticketing dataset. The work is organized into two connected stages: (i) a data engineering stage, in which raw transaction data is imported, cleaned, type-corrected, enriched with derived time-based features, and loaded into a MySQL database; and (ii) an analytical stage, in which the enriched dataset is queried back from the database and subjected to systematic exploratory analysis using statistical summaries and visualizations.

1.1 Objectives of the Study

- To quantify hourly variation in ticket sales and redemption volumes and identify the peak and off-peak time windows.

- To examine daily (day-of-week) demand patterns and determine which days generate the highest ticket activity.
- To analyze seasonal demand variation across Winter, Spring, Summer, and Fall.
- To assess the relationship between ticket sales and ticket redemptions and quantify any gap between the two.
- To smooth short-term fluctuations using rolling averages and surface the underlying demand trend.
- To translate the resulting insights into practical recommendations for scheduling and resource allocation.

1.2 Scope

The analysis is restricted to the four raw fields available in the source data — a unique transaction identifier, a timestamp, a redemption count, and a sales count — together with features engineered from the timestamp. No external variables (such as weather, ticket pricing, or marketing campaigns) were available and are therefore outside the scope of this paper; they are discussed as directions for future work in Section 7.

2. Dataset Description

The source dataset [5], 'Toronto Island Ferry Tickets', was supplied as a comma-separated values (CSV) file and loaded into a Pandas DataFrame for processing. The raw dataset consists of 261,538 rows and 4 columns, each row representing a fifteen-minute interval snapshot of ticketing activity.

Column (raw)	Description	Data Type
_id	Unique row / transaction identifier	int64
Timestamp	Date and time of the 15-minute interval record	object (string) → datetime64
Redemption Count	Number of tickets redeemed in the interval	int64
Sales Count	Number of tickets sold in the interval	int64

Table 1. Raw dataset schema as loaded from the source CSV file.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

2.1 Descriptive Statistics

A statistical summary of the numeric fields (computed via df.describe()) shows a right-skewed distribution for both sales and redemption counts — the mean substantially exceeds the median, indicating that most 15-minute intervals record low ticket activity while a smaller number of intervals (corresponding to peak hours) record very high activity.

Statistic	Redemption Count	Sales Count
Count	261,538	261,538
Mean	48.89	49.60
Std. Deviation	104.55	99.86
Minimum	0	0
25th Percentile	3	3
Median (50%)	11	13
75th Percentile	40	48
Maximum	7,216	7,229

Table 2. Descriptive statistics of the raw Redemption Count and Sales Count fields.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

3. Methodology

The methodology followed a standard EDA pipeline consisting of environment setup, data ingestion, cleaning, type correction, feature engineering, database persistence, and finally, visual and statistical analysis. All processing was performed in Python using Jupyter Notebooks (main.ipynb for data preparation and analytic.ipynb for the analytical stage).

3.1 Tools and Libraries

- NumPy [4] and Pandas [1] — numerical computation and tabular data manipulation.
- Matplotlib [2] — used directly for the pie charts (Figures 3, 4, 7) and the rolling-average line plot (Figure 8).
- Seaborn [3] (built on top of Matplotlib) — used specifically for the grouped and hue-coded bar chart visualizations (Figures 1, 2, 5, 6).
- SQLAlchemy [6] and mysql-connector — used to connect to and write the processed data into a MySQL database.

3.2 Data Cleaning

Column names were converted to lower-case with underscores in place of spaces for consistency (e.g., 'Redemption Count' → 'redemption_count'), and the identifier column was renamed from '_id' to 'id'. The Timestamp field was converted from a plain string (object dtype) to a proper datetime64 type using pd.to_datetime(), enabling downstream extraction of hour, day, month, and season components.

The cleaned dataset was systematically checked for data quality issues:

- Missing values: df.isnull().sum() returned zero missing values across every column.
- Duplicate rows: df.duplicated().sum() confirmed zero duplicate records.

- Negative values: both sales_count and redemption_count were checked for negative entries; none were found, confirming the counts are logically valid.

This confirms that the dataset arrived in a clean state, with no null values, no duplicated rows, and no negative (invalid) counts, requiring no imputation or row removal.

3.3 Feature Engineering

To enable temporal analysis, the following derived features were engineered from the timestamp column:

Derived Feature	Description	Example Value
hours	Hour of day extracted from timestamp (0–23)	22
day_name	Full name of the day of the week	Sunday
month_name	Full name of the month	December
month	Numeric month (1–12)	12
season	Meteorological season mapped from month	Winter
week_part	Weekday vs. Weekend classification	Weekend
day_period	Six-bucket period of day (Mid Night, Early Morning, Morning, Afternoon, Evening, Night, Late Night)	Late Night
gap	sales_count minus redemption_count (unredeemed ticket gap)	2

Table 3. Engineered features derived from the raw timestamp column.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

The season mapping followed the standard meteorological convention: December, January, and February as Winter; March, April, and May as Spring; June, July, and August as Summer; and September, October, and November as Fall. The day_period feature further subdivided each 24-hour day into six intuitive buckets (Mid Night: 12 AM–4 AM, Early Morning: 5 AM–8 AM, Morning: 9 AM–11 AM, Afternoon: 12 PM–3 PM, Evening: 4 PM–6 PM, Night: 7 PM–9 PM, Late Night: 10 PM–11 PM) to support more granular hourly-trend visualizations.

3.4 Database Persistence

After cleaning and enrichment, the processed DataFrame (261,538 rows × 11 columns) was written to a MySQL database named island_db using SQLAlchemy's create_engine and DataFrame.to_sql(), creating a table named ticket_trackingdata. The analytical notebook subsequently re-queried this table using pd.read_sql('SELECT * FROM ticket_trackingdata', ...), decoupling the data preparation stage from the analysis stage and mirroring a realistic production analytics workflow.

4. Exploratory Data Analysis

With a clean, feature-enriched dataset available in the database, the analytical stage examined ticket demand from five complementary angles: hourly trends, day-of-week trends, seasonal trends, the overall sales-versus-redemption balance, and a rolling-average smoothed trend. Each subsection below presents the underlying aggregation, the corresponding visualization, and the resulting interpretation.

4.1 Hourly Demand Trends

Ticket sales were aggregated by hour of day and day-period bucket to reveal the intraday demand curve.

Day Period	Hour Range	Approx. Total Sales (summed)	Demand Level
Mid Night	0–4	100,976	Off-Peak
Early Morning	5–8	569,867	Off-Peak
Morning	9–11	3,226,595	Moderate–High
Afternoon	12–15	5,376,397	Peak
Evening	16–18	2,404,351	Moderate
Night	19–21	958,202	Low
Late Night	22–23	335,663	Off-Peak

Table 4. Total ticket sales aggregated by day-period bucket, with corresponding demand level classification.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

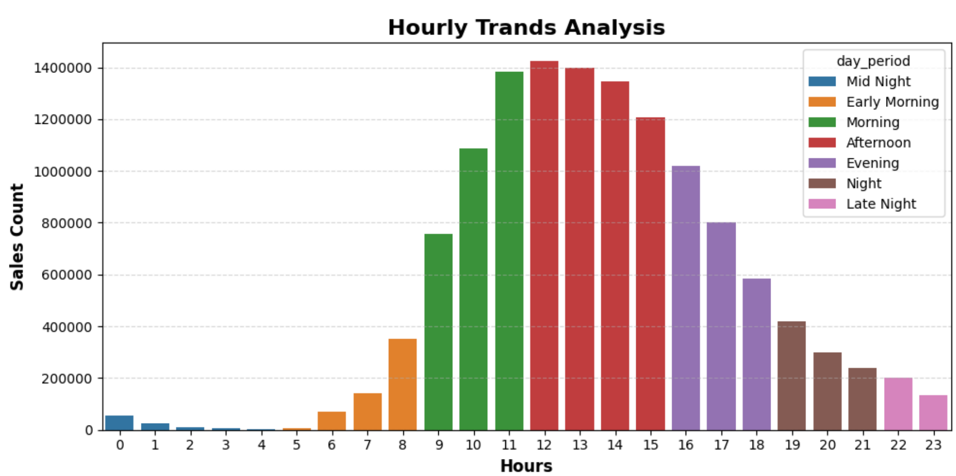

Figure 1. Hourly ticket sales by day-period, showing a pronounced afternoon peak.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

The hourly demand trend clearly shows that the Afternoon is the peak time for ticket sales, with the highest volumes recorded at 12:00 PM, closely followed by 1:00 PM, 2:00 PM, and 3:00 PM. Morning hours (9:00–11:00 AM) also show relatively high demand as activity ramps up ahead of the midday peak. In sharp contrast, ticket sales during Late Night and Mid Night hours are minimal, with almost no customer activity. The overall pattern is consistent with a leisure/tourism-driven ferry service where visitors predominantly travel in the middle of the day.

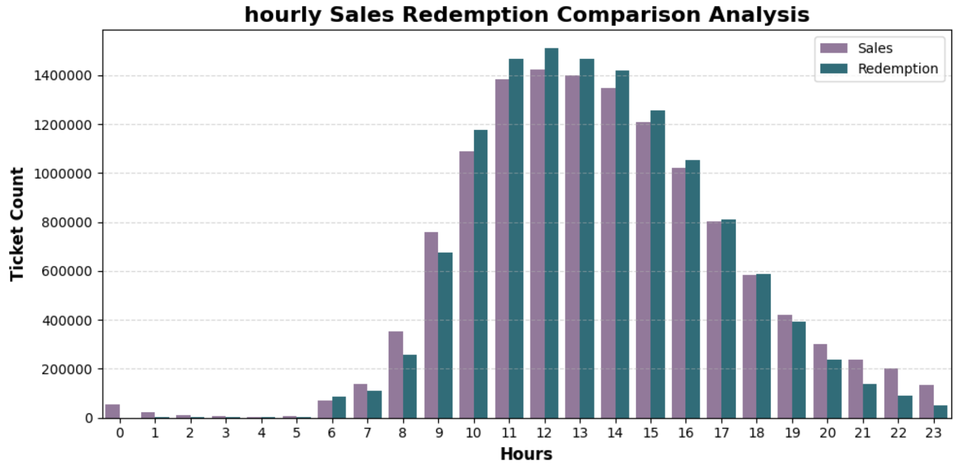

Figure 2. Side-by-side comparison of hourly ticket sales and ticket redemptions.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

- Both ticket sales and ticket redemptions are very low during the late-night hours, from 12:00 AM to 5:00 AM, reflecting minimal customer activity.
- Customer activity begins increasing from 6:00–7:00 AM, when redemptions start to rise ahead of sales — consistent with early travelers redeeming pre-purchased tickets.
- From around 9:00 AM, ticket sales also begin climbing, and both metrics grow together through the late morning.
- The peak hours for both sales and redemptions fall between 12:00 PM and 3:00 PM.
- After the afternoon peak, both sales and redemptions decline gradually through the evening and into the night.
- Overall, the afternoon is confirmed as the busiest period of the day, while late night and early morning represent the least busy hours.

4.2 Daily Demand Trends — Ticket Sales

Day	Total Sales	Share of Weekly Sales
Saturday	2,727,394	21.0%
Sunday	2,550,994	19.7%
Friday	1,784,931	13.8%
Monday	1,670,609	12.9%
Wednesday	1,444,624	11.1%
Thursday	1,461,639	11.3%
Tuesday	1,331,860	10.3%

Table 5. Total ticket sales by day of the week, ranked highest to lowest.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

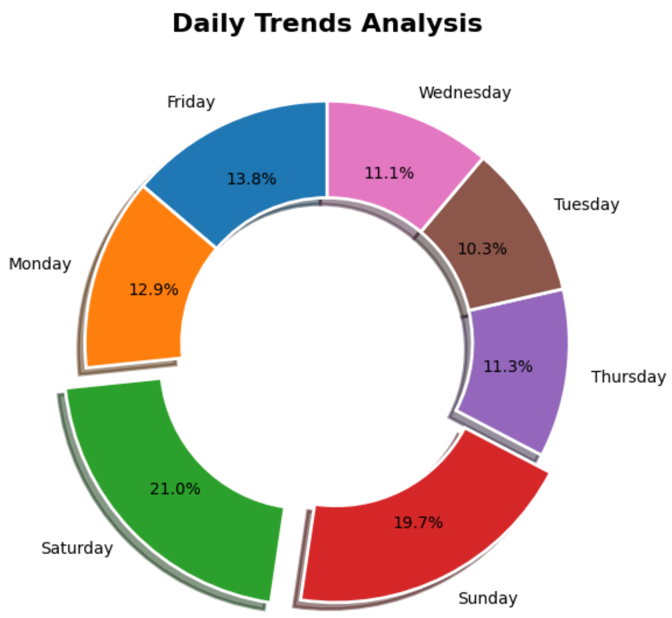

Figure 3. Distribution of ticket sales by day of the week.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

The weekend clearly dominates ticket sales: Saturday holds the largest share at 21.0%, followed by Sunday at 19.7%, together making the weekend the peak demand period. Among weekdays, Friday (13.8%) and Monday (12.9%) — days adjacent to the weekend — show comparatively higher demand than the mid-week. The lowest sales occur on Tuesday (10.3%), Wednesday (11.1%), and Thursday (11.3%), which together form the off-peak core of the working week.

4.3 Daily Demand Trends — Ticket Redemptions

Day	Total Redemptions
Saturday	2,795,008
Sunday	2,681,726
Friday	1,621,026
Monday	1,654,352
Wednesday	1,383,026
Thursday	1,379,066
Tuesday	1,271,089

Table 6. Total ticket redemptions by day of the week.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

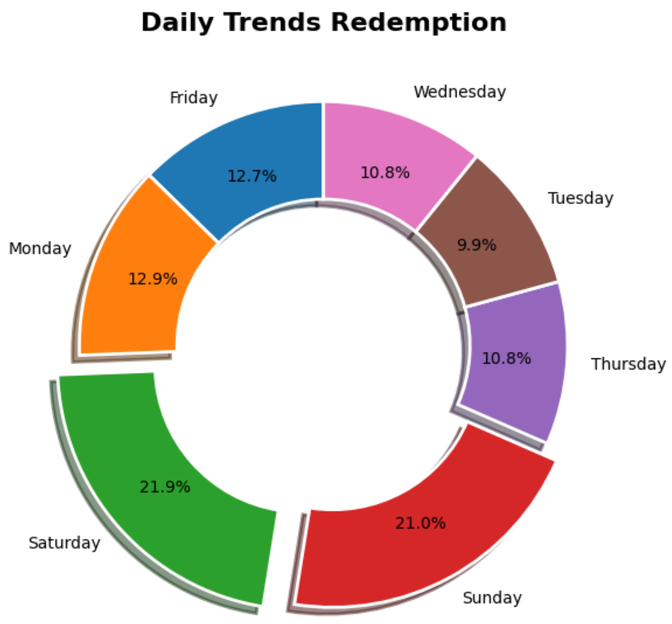

Figure 4. Distribution of ticket redemptions by day of the week.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

The redemption pattern mirrors the sales pattern almost exactly, with Saturday and Sunday again accounting for the largest shares. Notably, on both weekend days redemption volumes exceed sales volumes, indicating that a portion of tickets pre-purchased on adjoining days are redeemed during the weekend rush — a sign of heavy weekend footfall relative to ticket purchase timing.

4.4 Sales vs. Redemption Comparison by Day

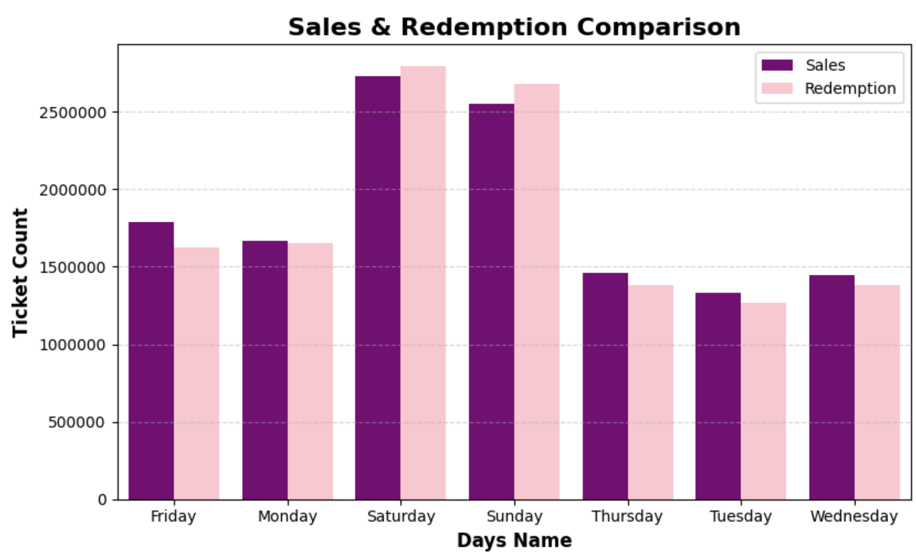

Figure 5. Grouped bar comparison of ticket sales and redemptions across all seven days.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

- There is only a small difference between ticket sales and ticket redemptions on every day of the week, indicating that most sold tickets are eventually redeemed.

- Saturday and Sunday record the highest values for both sales and redemptions, confirming them as the peak business days.
- On Saturday and Sunday specifically, redemption count is slightly higher than sales count, pointing to very high customer activity and same-day usage during the weekend.
- Tuesday, Wednesday, and Thursday show the lowest sales and lowest redemptions, marking them as the least busy days of the week.
- A strong positive relationship exists between the two metrics — as sales rise, redemptions rise in tandem, and vice versa.
- Overall, the weekend generates the highest customer demand, while the middle of the week experiences comparatively low activity.

4.5 Seasonal Comparison

Season	Total Sales	Total Redemptions
Summer	8,341,015	8,000,961
Fall	2,498,657	2,606,380
Spring	1,693,899	1,664,773
Winter	438,480	513,179

Table 7. Total ticket sales and redemptions grouped by season.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

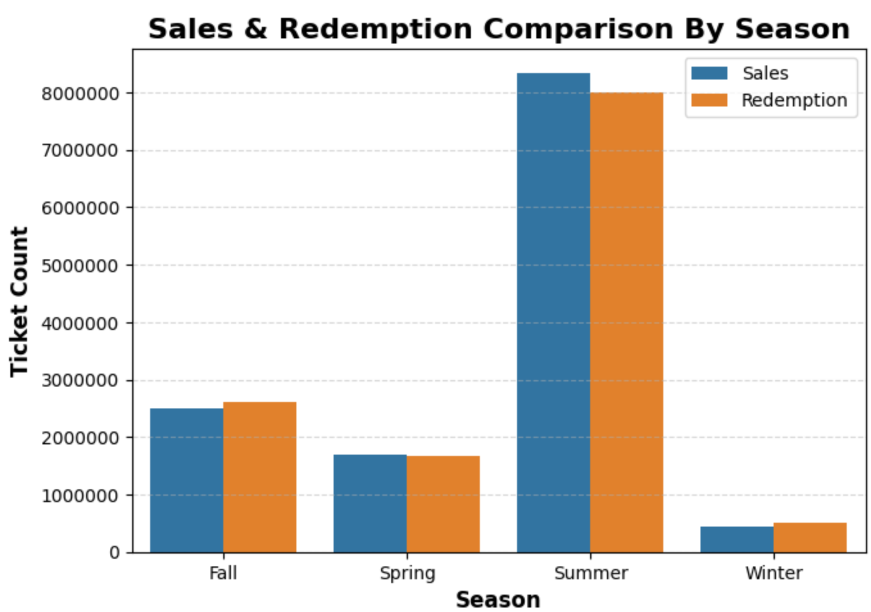

Figure 6. Seasonal comparison of ticket sales and redemptions.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

Summer emerges as the peak season by a substantial margin, accounting for roughly 60% of total annual sales volume in this dataset — far outstripping the other three seasons combined. This aligns with expectations for a recreational island ferry service, where warm-weather months drive the bulk of tourism and outdoor activity. Ticket redemptions in Summer are slightly lower than sales, though the difference is minor. Winter, conversely, is the clear off-peak season, recording the lowest sales and redemptions of the year; interestingly, redemptions slightly exceed sales in Winter, though overall volumes remain very low. Spring and Fall occupy a moderate middle ground, showing neither a pronounced peak nor a pronounced trough.

4.6 Sales vs. Redemption Distribution (Overall)

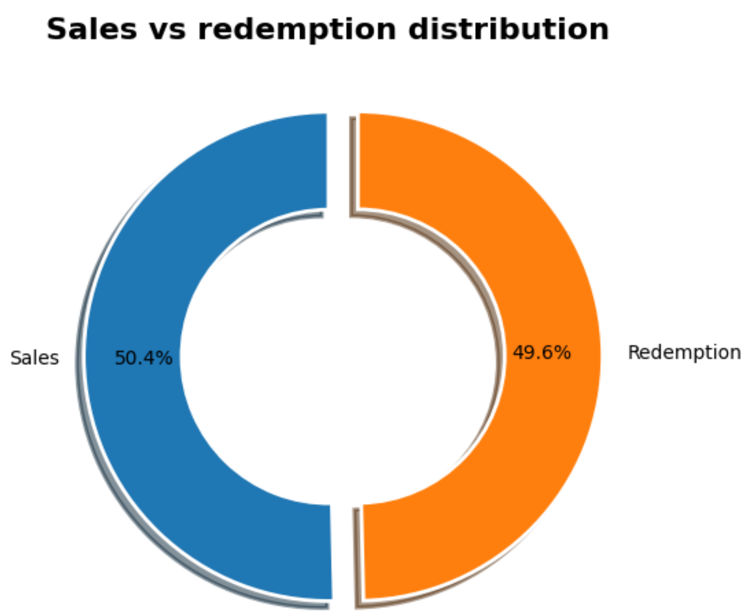

Figure 7. Overall distribution of total ticket sales versus total ticket redemptions across the full dataset.
Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

- Ticket sales account for 50.4% of the combined sales-plus-redemption volume, while ticket redemptions account for 49.6%.
- The difference between the two is only 0.8 percentage points — a very small gap.
- This indicates that almost all sold tickets are eventually redeemed by customers.
- The near-equal split reflects a strong balance between ticket sales and ticket redemptions across the entire observation period.
- Overall, this confirms efficient ticket usage with minimal wastage or unredeemed inventory.

4.7 Rolling Average Trend Analysis

To separate the underlying demand signal from short-term hour-to-hour noise, 1-hour and 4-hour rolling averages were computed on the mean hourly sales_count series.

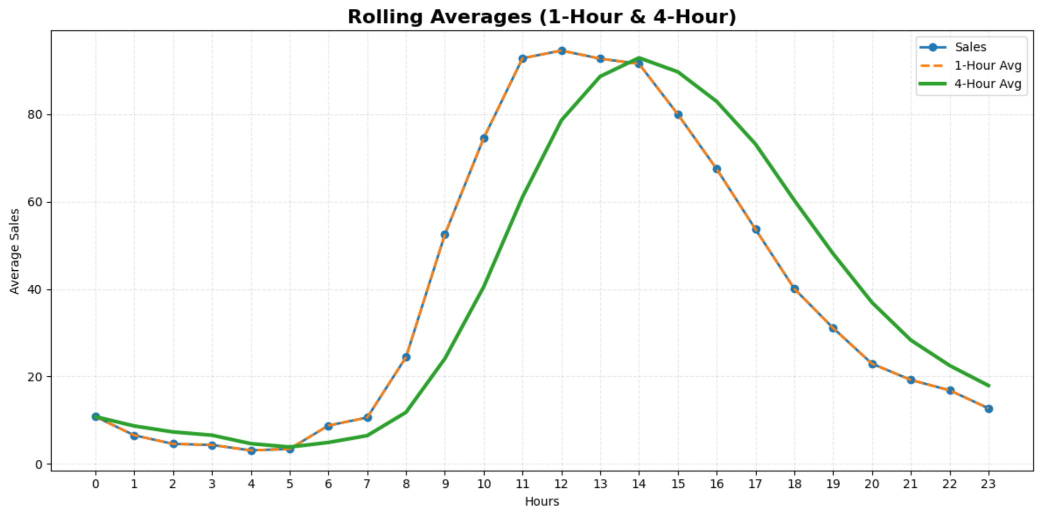

Figure 8. Original hourly sales overlaid with 1-hour and 4-hour rolling averages.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

- The 1-hour rolling average tracks the original hourly sales figures almost exactly, since it is computed from a single hour of data at a time.
- The 4-hour rolling average is visibly smoother and more clearly exposes the overall daily sales trend by damping short-term fluctuations.
- Sales remain very low during the early morning hours, from roughly 12 AM to 6 AM.
- Sales begin increasing after 7 AM and rise rapidly through the morning.
- The highest sales are recorded between 11 AM and 2 PM, confirming this as the peak business period identified throughout the analysis.
- After approximately 2 PM, sales gradually decline through the evening and into the night.
- The 4-hour rolling average removes sudden spikes and dips, making the single-peak, gradually-declining nature of the daily demand curve unambiguous.

4.8 Interactive Dashboard Overview

To make the findings from Sections 4.1 to 4.7 easy to explore, an interactive dashboard was built on top of the same enriched dataset. The dashboard lets a user filter by date, month, hour, peak/off-peak status, and season, and it updates all charts and KPI cards live based on the selected filters.

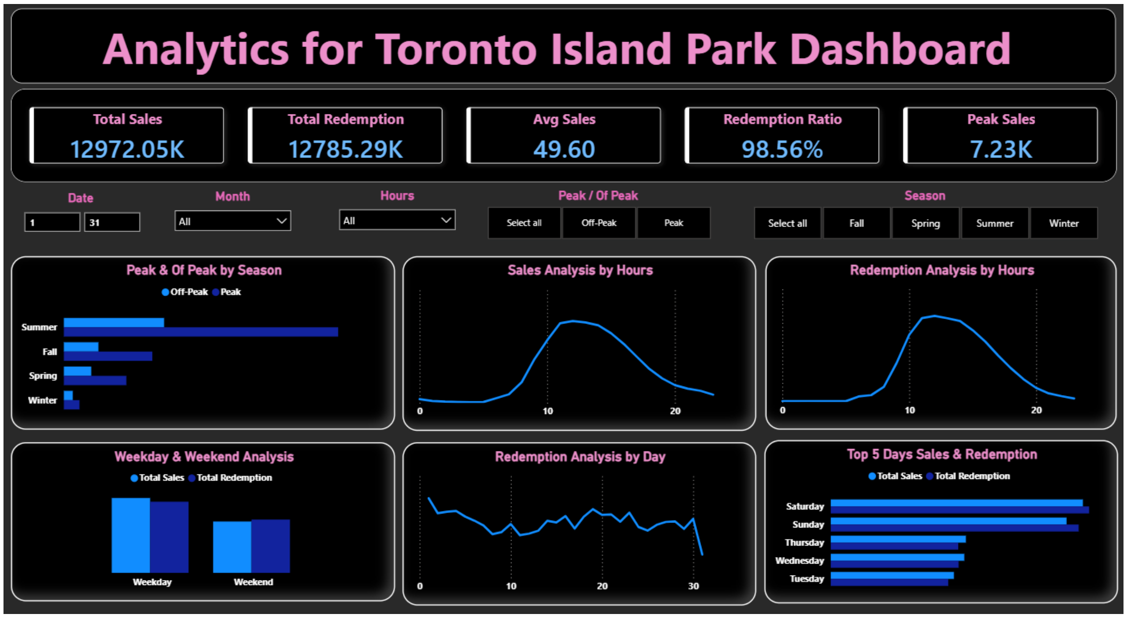

Figure 9. Interactive dashboard for Toronto Island Park ferry ticket sales and redemption analytics.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

Simple Insights from the Dashboard

- Total Sales are about 12,972.05K and Total Redemption is about 12,785.29K, so almost every sold ticket gets used.
- Average Sales per record is 49.60, matching the mean value found earlier in the descriptive statistics (Table 2).
- Redemption Ratio is 98.56%, which means out of every 100 tickets sold, close to 99 are actually redeemed.
- Peak Sales touch 7.23K, showing the single busiest moment recorded in the data.
- The 'Peak & Off-Peak by Season' chart shows Summer has by far the longest Peak bar, confirming Summer as the busiest season.
- The 'Sales Analysis by Hours' and 'Redemption Analysis by Hours' line charts both rise in the middle of the day and fall at night, which matches the afternoon peak already found in Section 4.1.
- The 'Weekday & Weekend Analysis' chart shows Weekend bars are taller than Weekday bars for both sales and redemption, confirming weekends are busier.
- The 'Top 5 Days Sales & Redemption' chart lists Saturday and Sunday at the top, followed by Thursday, Wednesday, and Tuesday, which lines up with the day-of-week findings in Section 4.2 and 4.3.
- Overall, the dashboard visually confirms the same story as the written analysis: demand is highest in the afternoon, on weekends, and in Summer, and very few sold tickets go unredeemed.

5. Summary of Key Findings

#	Dimension	Key Finding
1	Hourly	Demand peaks between 12:00 PM and 3:00 PM (Afternoon); lowest activity occurs from 12:00 AM to 5:00 AM.
2	Daily	Saturday (21.0%) and Sunday (19.7%) together account for over 40% of weekly ticket sales; Tuesday is the quietest day (10.3%).
3	Weekly Balance	Sales and redemptions move almost in lockstep across all seven days, with weekend redemptions slightly exceeding sales.
4	Seasonal	Summer dominates annual demand (~60% of yearly sales); Winter is the clear off-peak season.
5	Sales vs. Redemption	Overall split is 50.4% sales to 49.6% redemptions — only a 0.8-point gap, indicating minimal unredeemed ticket leakage.
6	Trend Shape	A single, well-defined midday demand peak exists; the 4-hour rolling average confirms no secondary spikes elsewhere in the day.

Table 8. Consolidated summary of key EDA findings across all analytical dimensions.

Source: Author's analysis, based on the Toronto Island Ferry Tickets dataset.

6. Conclusion and Recommendations

This exploratory data analysis of 261,538 ferry ticketing transactions reveals a highly predictable, structured demand pattern for the Toronto Island Park ferry service. Demand is concentrated in a well-defined afternoon window, is markedly higher on weekends than weekdays, and peaks strongly during the Summer season. The close statistical correspondence between sales and redemption volumes indicates that the current ticketing process is operationally efficient, with a minimal gap between tickets purchased and tickets actually used.

Based on these findings, the following recommendations are proposed for operational planning:

- Staffing and vessel capacity should be scaled up during the 12:00 PM–3:00 PM afternoon window on all days, and especially on Saturdays and Sundays, when combined demand is highest.
- Off-peak periods — late night, early morning, and weekday mid-week (Tuesday–Thursday) — present opportunities for reduced staffing, scheduled maintenance windows, or promotional pricing to smooth demand.

- Seasonal workforce and fleet planning should anticipate a large Summer surge (roughly six times Winter volumes) and scale resources accordingly, while Winter operations can run on a leaner schedule.
- Because redemption counts slightly exceed sales counts on weekends, ticket validity windows and pre-sale policies could be reviewed to ensure capacity is not over-committed during the busiest days.
- The 4-hour rolling average pattern can serve as a baseline for automated, near-real-time demand alerts that flag unusual deviations from the expected midday peak.

7. Limitations and Future Work

This study is subject to several limitations that also point toward directions for future research.

- The dataset contains only four raw variables (id, timestamp, sales count, redemption count); external factors such as weather conditions, public holidays, special events, and ticket pricing were not available and could meaningfully influence demand.
- The analysis is descriptive rather than predictive; no forecasting model (e.g., ARIMA, Prophet, or a machine learning regressor) was built to project future demand, though the rolling-average work in Section 4.7 provides a natural foundation for such a model.
- The season and day_period bucketing schemes used fixed, rule-based definitions; a data-driven clustering approach could potentially reveal more nuanced demand regimes.
- The dataset's exact date range and year coverage were not independently verified beyond the sample timestamps observed; a fuller multi-year dataset would strengthen the seasonal conclusions.
- Future work could incorporate weather and holiday-calendar data, build a short-term demand forecasting model, and develop an automated anomaly-detection layer on top of the rolling-average baseline established here.
- Formal statistical hypothesis testing (e.g., ANOVA or the Kruskal-Wallis test) could be applied to verify whether the observed differences across days of the week and seasons are statistically significant, rather than relying on descriptive comparisons alone.
- Where available and privacy-compliant, passenger and ticket-type segmentation data (e.g., single-ride versus multi-day passes, resident versus tourist status) could reveal demand drivers that are not observable from the aggregate sales and redemption counts used in this study.

8. References

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[2] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90–95.

[3] Waskom, M. L. (2021). Seaborn: Statistical Data Visualization. Journal of Open Source Software, 6(60), 3021.

[4] Harris, C. R. et al. (2020). Array Programming with NumPy. Nature, 585, 357–362.

[5] Toronto Island Ferry Tickets Dataset (source CSV file, as supplied for this analysis: 'Toronto Island Ferry Tickets(1).csv').

[6] SQLAlchemy Documentation — https://docs.sqlalchemy.org/ (accessed for MySQL database connectivity implementation).